

“Analysis of Spatial Smoothing with Uniform Circular Arrays”

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Overview of Paper

- This paper [1] analyses the asymptotic performance of **Root-MUSIC** using i) **Forward** smoothing and ii) **Forward-Backward** smoothing in mode-space with:
 - Correlated sources
 - Finite data perturbations (finite snapshots)
 - Beam-space transform approximation error
- Incorporating **directional** elements

Beamspace Transform Error

- When transforming UCA manifold to virtual ULA manifold, we neglect the residual terms in the expansion (valid for $d < 0.5\lambda$).

UCA Steering Vector:

$$a_c(\theta) = \left[e^{j\xi \cos \theta}, e^{j\xi \cos(\theta - \frac{2\pi}{L})}, \dots, e^{j\xi \cos(\theta - \frac{2\pi(L-1)}{L})} \right]^T \quad (1)$$

Array Pattern (m -th phase mode):

$$f_m(\theta) = \underbrace{J_{|m|}(\xi)e^{jm\theta}}_{\text{Principal Term}} + \underbrace{j^{-|m|} \sum_{q=1}^{\infty} [j^q J_q(\xi)e^{-jq\theta} + j^q J_q(\xi)e^{jq\theta}]}_{\text{Residual Terms}} \quad (2)$$

Transformed Steering Vector:

$$a_t(\theta) = F^H a_c(\theta) = J_{\xi} a(\theta) + \underbrace{\Delta a(\theta)}_{\text{error term}} \quad (3)$$

where, $J_{\xi} = \sqrt{L} \text{diag} [J_D(\xi), \dots, J_1(\xi), J_0(\xi), J_1(\xi), \dots, J_D(\xi)]$

$$a(\theta) = \left[e^{-jD\theta}, e^{-j(D-1)\theta}, \dots, 1, \dots, e^{j(D-1)\theta}, e^{jD\theta} \right]^T$$

Error in transformed Covariance Matrix

UCA Covariance Matrix:

$$R_c = A_c S A_c^H + \sigma^2 I \quad (4)$$

where, S = signal covariance matrix

A_c = matrix of direction vectors of the UCA

I = identity matrix

Transformed Covariance Matrix:

$$R^t = F^H R_c F = J_\xi A S A^H J_\xi + \sigma^2 I + \underbrace{\Delta R}_{\text{Error Covariance}} \quad (5)$$

where, $\Delta R = \Delta A S A^H J_\xi + J_\xi A S \Delta A^H + \Delta A S \Delta A^H$

$$A = [a(\theta_1), \dots, a(\theta_M)]$$

Forward Smoothing and Prewhitening

- Spatial smoothing 'colours' the noise. We need pre-whitening post-transform and pre-MUSIC application.

Forward Smoothed Covariance Matrix:

$$R_f^t = \frac{1}{K} \sum_{l=1}^K Z_l^T J_\xi^{-1} R^t J_\xi^{-1} Z_l \quad (6)$$

where, $Z_l = [e_l, e_{l+1}, \dots, e_{l+L_o-1}]$

$L_o = 2D - K + 2$ (virtual subarray dimension)

K = number of virtual subarrays

Prewhitened Smoothed Covariance Matrix:

$$(R_f^t)_w = R_{nw} R_f^t R_{nw}^H \quad (7)$$

$$\text{where, } R_{nw} = \left[\frac{1}{K} \sum_{l=1}^K Z_l^T J_\xi^{-1} J_\xi^{-1} Z_l \right]^{-1/2}$$

R_{nw} = prewhitening matrix

Analysis of Forward Smoothing

- **Effective Correlation:** Neglecting the error covariance ΔR , the effective correlation coefficient post-smoothing is given by [2]:

$$|\rho_f| = \left| \rho \frac{\sin(K(\theta_i - \theta_j)/2)}{K \sin((\theta_i - \theta_j)/2)} \right| \quad (8)$$

- **Key Properties:**

- ρ_f depends *only* on the angular separation $(\theta_i - \theta_j)$, not on individual directions.
- It is completely independent of inter-element spacing.
- *Contrast:* In a ULA, correlation depends on both spacing and individual directions.

- **Error Analysis [3]:**

- While the full Mean Square Error (MSE) equation is highly complex, the crux is that as inter-element spacing $d \rightarrow 0.5\lambda$, the norm of the error vector approaches 0 dB.
- **Impact of Smoothing:** Spatial smoothing successfully reduces this overall error norm.
- **Limitation:** This analytical derivation of MSE holds only for uncorrelated (non-coherent) sources. For coherent sources, the authors note the expressions become intractably complex even for just two sources.

Impact of Spatial Smoothing on Error Norm

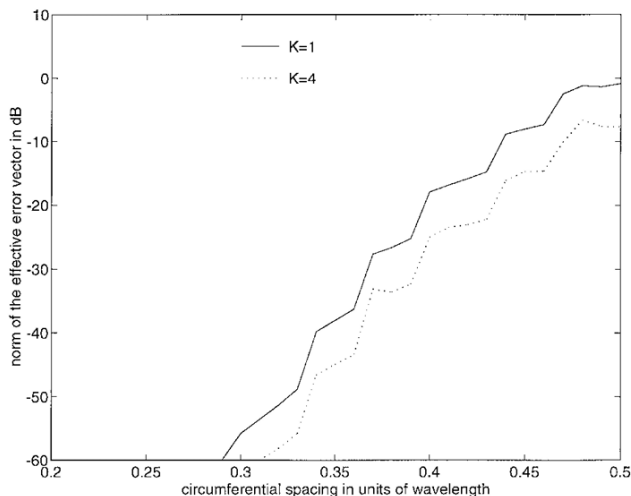


Figure: Norm of the effective error vector in the transformed steering vector of UCA as a function of circumferential spacing ($L = 30$).

Forward-Backward Spatial Smoothing

Forward-Backward Smoothed Covariance:

$$R_{fb}^t = \frac{1}{2K} \sum_{l=1}^K Z_l^T \left[J_\xi^{-1} R^t J_\xi^{-1} + \bar{I} (J_\xi^{-1} R^t J_\xi^{-1})^* \bar{I} \right] Z_l \quad (9)$$

where, $\bar{I}$ = exchange matrix

$(\cdot)^*$ = complex conjugate operator

Effective Correlation Coefficient:

$$|\rho_{fb}| = |\rho| \left| \frac{\sin(K(\theta_i - \theta_j)/2) \cos \psi}{K \sin((\theta_i - \theta_j)/2)} \right| \quad (10)$$

- **Key Insight:** Here, ψ is the phase angle of the complex correlation coefficient ($\rho = |\rho| e^{j\psi}$).
- If ρ is strictly real (i.e., $\psi = 0$), the performance of forward-backward smoothing reduces to be exactly the same as just forward smoothing.

Spatial Smoothing with Directional Elements

- **Directional Pattern:** Assuming identical sensors, the periodic directional pattern $G(\theta)$ can be expanded as a Fourier series [4]:

$$G(\theta) = \sum_{l=-Q}^Q c_l e^{jl\theta} \quad (11)$$

- **Transformed Steering Vector:** Applying the phase mode transformation yields a similar ULA-like structure:

$$F^H a_{cd}(\theta) = J_{d\xi} a(\theta) + \Delta a_d(\theta)$$

- **Key Takeaways:**
 - Since the ULA-like structure of $a(\theta)$ is preserved, Root-MUSIC and spatial smoothing can be successfully extended to UCA with directional elements.
 - **Crucial Difference:** The new diagonal matrix $J_{d\xi}$ is **complex** (unlike the real J_ξ for omnidirectional sensors).
 - Consequently, its complex conjugate $J_{d\xi}^*$ must be used wherever appropriate in the smoothing equations.

Simulation Insights & Key Results

- **The Smoothing “Sweet Spot”:** Increasing virtual subarrays (K) initially reduces MSE. However, excessive smoothing degrades performance due to a reduced effective aperture.
- **Source Correlation:** Smoothing improves DOA estimation overall, but performance gains are vastly superior for uncorrelated sources.
- **Directional Elements:** For closely spaced, correlated sources, Forward-Backward smoothing significantly outperforms standard Forward smoothing.
- **Widely Spaced Sources:** MSE is highly resilient to the specific choice of K , providing a wide operational margin for array design.

References

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